

MEMVERSE: Zero-Trust Privacy Gateway for Multimodal AI Memory

A client-governed security architecture reconciling personalization and data privacy in large language models.

1. Project Description & Core Highlights

MEMVERSE is a zero-trust privacy gateway designed to enable continuous, personalized AI interactions without exposing raw personal identifiers to external model providers. By enforcing a 12-stage security pipeline with dynamic policy transformations, cryptographic memory passports, and tamper-evident audit ledgers, MEMVERSE ensures that external LLMs receive only the minimum necessary generalized semantic context.

12-Stage Security Pipeline: Client-side PII masking, biometric blinding, and context minimization.

Cryptographic Audit Ledger: SHA-256 hash chaining guaranteeing immutable proof of compliance.

2. Team Details

Name	Role	Institution & Department
Satvik Kesarwani	Team Lead	Indian Institute of Information Technology (IIIT), Pune B.Tech, Computer Science and Engineering (MIS: 112315166)
Astha Jain	Core Developer	Institute of Engineering and Technology (IET DAVV), Indore

3. Project Deliverables & Submission Links

1. Project Video Demonstration (High Priority)

Walkthrough video recording showcasing zero-trust memory, multimodal document ingestion, biometric privacy, and live trace inspection:

<https://drive.google.com/file/d/1VjpQrxNEMAkfgZdqcdH9a4bNIQCHAY9H/view?usp=sharing>

2. Live Working Prototype — Frontend (High Priority)

Interactive web client featuring real-time chat, policy exploration, memory passport management, and 12-stage trace inspection:

<https://memverse-satvikkesarwanis-projects.vercel.app>

3. Gateway API Service (Backend)

FastAPI zero-trust gateway service integrated with live NVIDIA NIM models and cryptographic ledger endpoints:

<https://memverse-api.onrender.com/api/status>

4. Source Code Repository

Complete project source code repository including backend services, frontend application, and test suites:

<https://github.com/satvikkesarwani/memverse>

5. Project Details & Demonstration Media Folder

Submission repository containing project demonstration recordings, architecture diagrams, and supporting documentation:

https://drive.google.com/drive/folders/1U03jMmCsi0P9CiwQckT6S5NyeJYVo_f9?usp=sharing